Patient Record — ID 2051

Age	45	**Sex**	M
Race/Ethnicity	Non-Hispanic White	**Insurance**	Medicare primary (ESRD eligibility), Blue Cross Blue Sh
Total Encounters	8	**Generation Seed**	5240

Primary Diagnoses

• Acute rejection of lung transplant
• Chronic kidney allograft rejection
• Autosomal Dominant Polycystic Kidney Disease
• History of Pneumocystis jirovecii pneumonia
• Diabetic ketoacidosis
• Cytomegalovirus colitis
• Post-transplant lymphoproliferative disorder, polymorphic type, EBV-associated

Comorbidities

• History of kidney transplantation
• Iatrogenic immunosuppression
• End-stage renal disease on hemodialysis
• Essential hypertension
• Hyperkalemia

Patient Profile

Sex	M
Age at First Encounter	45
Race/Ethnicity	Non-Hispanic White
Insurance	Medicare primary (ESRD eligibility), Blue Cross Blue Shield secondary
Occupation	High school mathematics teacher
Smoking Status	Former smoker, quit 12 months prior to transplantation
Alcohol Use	Social (1-2 drinks per month)

Allergies:

• NKDA

Chronic Conditions:

• Autosomal dominant polycystic kidney disease
• Status post deceased donor renal transplantation
• Type 2 diabetes mellitus
• Hypertension
• Chronic kidney disease stage 5T

Home Medications:

• Tacrolimus - 5 mg PO BID (target trough 8-10 ng/mL)
• Mycophenolate mofetil - 750 mg PO BID

• Prednisone - 5 mg PO daily
• Insulin glargine - 24 units subcutaneous at bedtime
• Insulin lispro - Sliding scale 1:12 ratio with meals
• Amlodipine - 10 mg PO daily
• Trimethoprim-sulfamethoxazole - Single-strength tablet PO daily (PCP prophylaxis)
• Aspirin - 81 mg PO daily

Surgical History:

• Deceased donor renal transplantation (6 months prior to initial encounter)
• Left native nephrectomy (10 years prior for infected cyst)
• Right native kidney in situ

Family History:

• Father: Autosomal dominant polycystic kidney disease, deceased age 58 from ruptured cerebral aneurysm
• Mother: Hypertension, type 2 diabetes mellitus
• Sister: Autosomal dominant polycystic kidney disease, currently on hemodialysis

Encounters (8)

Encounter 1 — 2020-01-15 — Nephrology

Encounter ID	8617	**Type**	inpatient
Attending	Dr. Sarah Chen	**Order**	0
Department	Nephrology	**Gen. Method**	template

Structured EHR Sections:

CHIEF_COMPLAINT (modified)

Decreased urine output, fatigue, and lower extremity swelling two weeks after kidney transplantation

HPI

Two weeks after receiving a renal transplant

PMH (modified)

Active Problem List:
- Autosomal dominant polycystic kidney disease
- Status post deceased donor renal transplantation
- Type 2 diabetes mellitus
- Hypertension
- Chronic kidney disease stage 5T

PSH (modified)

- Deceased donor renal transplantation (6 months prior to initial encounter)
- Left native nephrectomy (10 years prior for infected cyst)
- Right native kidney in situ

MEDICATIONS (modified)

Home Medications:
- Tacrolimus 5 mg PO BID (target trough 8-10 ng/mL)
- Mycophenolate mofetil 750 mg PO BID
- Prednisone 5 mg PO daily
- Insulin glargine 24 units subcutaneous at bedtime
- Insulin lispro Sliding scale 1:12 ratio with meals
- Amlodipine 10 mg PO daily
- Trimethoprim-sulfamethoxazole Single-strength tablet PO daily (PCP prophylaxis)
- Aspirin 81 mg PO daily

ALLERGIES (modified)

NKDA

FAMILY_HISTORY (modified)

- Father: Autosomal dominant polycystic kidney disease, deceased age 58 from ruptured cerebral aneurysm
- Mother: Hypertension, type 2 diabetes mellitus
- Sister: Autosomal dominant polycystic kidney disease, currently on hemodialysis

SOCIAL_HISTORY (modified)

Smoking: Former smoker, quit 12 months prior to transplantation. Alcohol: Social (1-2 drinks per month). Occupation: High school mathematics teacher.

VITALS

Blood pressure of 160/95 mm Hg and a temperature of 37.2°C (99°F)

PHYSICAL_EXAM

Mild bilateral lower extremity edema and left lower quadrant tenderness near the graft site

LABS

Serum creatinine of 3.2 mg/dL (baseline 1.1 mg/dL), BUN of 46 mg/dL, and elevated serum potassium. Urinalysis shows mild proteinuria and white blood cell casts.

PATHOLOGY

Biopsy of the renal graft demonstrates mononuclear cell infiltration and tubular injury

Encounter 2 — 2020-03-15 — Emergency Medicine

Encounter ID	8620	**Type**	ed
Attending	Dr. Marcus Johnson	**Order**	1
Department	Emergency Medicine	**Gen. Method**	template

Structured EHR Sections:

CHIEF_COMPLAINT (modified)

Worsening right flank pain and gross hematuria

PMH (modified)

Active Problem List:
- Acute rejection of lung transplant (diagnosed 2020-01-15)
- Autosomal dominant polycystic kidney disease
- Status post deceased donor renal transplantation
- Type 2 diabetes mellitus
- Hypertension
- Chronic kidney disease stage 5T

PSH (modified)

- Deceased donor renal transplantation (6 months prior to initial encounter)
- Left native nephrectomy (10 years prior for infected cyst)
- Right native kidney in situ

MEDICATIONS (modified)

Home Medications:
- Tacrolimus 5 mg PO BID (target trough 8-10 ng/mL)
- Mycophenolate mofetil 750 mg PO BID
- Prednisone 5 mg PO daily
- Insulin glargine 24 units subcutaneous at bedtime
- Insulin lispro Sliding scale 1:12 ratio with meals
- Amlodipine 10 mg PO daily
- Trimethoprim-sulfamethoxazole Single-strength tablet PO daily (PCP prophylaxis)
- Aspirin 81 mg PO daily

ALLERGIES (modified)

NKDA

FAMILY_HISTORY

He has a family history of kidney disease; his father required dialysis in his 50s.

SOCIAL_HISTORY (modified)

Smoking: Former smoker, quit 12 months prior to transplantation. Alcohol: Social (1-2 drinks per month). Occupation: High school mathematics teacher.

VITALS

Blood pressure is 160/100 mmHg

PHYSICAL_EXAM

Physical examination shows bilateral flank masses

LABS

Urinalysis shows microscopic hematuria and mild proteinuria

IMAGING

A renal ultrasound demonstrates multiple large cysts in both kidneys. Brain MRI performed for headache reveals a saccular aneurysm at the anterior communicating artery.

ASSESSMENT

A mutation involving

PLAN

Genetic testing is ordered.

Encounter 3 — 2020-04-15 — Gastroenterology

Encounter ID	8623	**Type**	inpatient
Attending	Dr. Priya Patel	**Order**	2
Department	Gastroenterology	**Gen. Method**	hybrid

Structured EHR Sections:

CHIEF_COMPLAINT (modified)

Abdominal pain, vomiting, and bloody diarrhea

HPI (modified)

The patient with ADPKD s/p renal transplant, prior rejection, and hematuria presents to the clinic with a 1-week history of abdominal pain, vomiting, and bloody diarrhea. He denies any recent travel or sick contacts.

PMH (modified)

Active Problem List:
- Acute rejection of lung transplant (diagnosed 2020-01-15)
- Autosomal Dominant Polycystic Kidney Disease (diagnosed 2020-03-15)
- Status post deceased donor renal transplantation
- Type 2 diabetes mellitus
- Hypertension
- Chronic kidney disease stage 5T

PSH

He underwent a renal transplantation 3 months ago

MEDICATIONS (modified)

He has been on immunosuppressive therapy with tacrolimus, mycophenolate mofetil, and prednisone.

Home Medications:
- Insulin glargine 24 units subcutaneous at bedtime
- Insulin lispro Sliding scale 1:12 ratio with meals
- Amlodipine 10 mg PO daily
- Trimethoprim-sulfamethoxazole Single-strength tablet PO daily (PCP prophylaxis)
- Aspirin 81 mg PO daily

ALLERGIES (modified)

NKDA

FAMILY_HISTORY (modified)

- Father: Autosomal dominant polycystic kidney disease, deceased age 58 from ruptured cerebral aneurysm
- Mother: Hypertension, type 2 diabetes mellitus
- Sister: Autosomal dominant polycystic kidney disease, currently on hemodialysis

SOCIAL_HISTORY (modified)

Smoking: Former smoker, quit 12 months prior to transplantation. Alcohol: Social (1-2 drinks per month). Occupation: High school mathematics teacher.

VITALS

On examination, he is afebrile, blood pressure is 130/85 mmHg, heart rate is 88 beats per minute

PHYSICAL_EXAM

Abdominal examination reveals diffuse tenderness without rebound or guarding.

LABS

Laboratory tests reveal the following:

- Hemoglobin: 12.1 g/dL
- White blood cell count: 9,000/uL with a normal differential
- Platelets: 250,000/uL
- Serum creatinine: 1.2 mg/dL (baseline post-transplant)
- Liver enzymes: within normal limits
- CMV PCR: Positive with high viral load

OTHER_STUDIES

An endoscopy is performed, showing multiple, large, shallow ulcers in the colon. Biopsies of the ulcers are taken.

Encounter 4 — 2020-05-15 — Nephrology

Encounter ID	8618	**Type**	outpatient
Attending	Dr. James Wilson	**Order**	3
Department	Nephrology	**Gen. Method**	hybrid

Structured EHR Sections:

CHIEF_COMPLAINT (modified)

Routine follow-up for rising serum creatinine

HPI (modified)

This 45-year-old male with autosomal dominant polycystic kidney disease returns for routine nephrology follow-up four months after deceased donor kidney transplantation, following cytomegalovirus colitis one month ago. Four months after receiving a kidney transplant due to end-stage renal disease secondary to hypertension, he has no complaints and has been adherent to his immunosuppressive regimen, which includes tacrolimus, mycophenolate mofetil, and prednisone. The patient's serum creatinine level has risen from a baseline of 1.2 mg/dL to 2.8 mg/dL since his last visit one month ago.

PMH (modified)

Active Problem List:
- Acute rejection of lung transplant (diagnosed 2020-01-15)
- Autosomal Dominant Polycystic Kidney Disease (diagnosed 2020-03-15)
- Cytomegalovirus colitis (diagnosed 2020-04-15)
- Status post deceased donor renal transplantation
- Type 2 diabetes mellitus
- Hypertension
- Chronic kidney disease stage 5T

PSH (modified)

- Deceased donor renal transplantation (6 months prior to initial encounter)
- Left native nephrectomy (10 years prior for infected cyst)
- Right native kidney in situ

MEDICATIONS (modified)

He takes tacrolimus, mycophenolate mofetil, and prednisone.

Home Medications:
- Insulin glargine 24 units subcutaneous at bedtime
- Insulin lispro Sliding scale 1:12 ratio with meals
- Amlodipine 10 mg PO daily
- Trimethoprim-sulfamethoxazole Single-strength tablet PO daily (PCP prophylaxis)
- Aspirin 81 mg PO daily

ALLERGIES (modified)

NKDA

FAMILY_HISTORY (modified)

- Father: Autosomal dominant polycystic kidney disease, deceased age 58 from ruptured cerebral aneurysm
- Mother: Hypertension, type 2 diabetes mellitus
- Sister: Autosomal dominant polycystic kidney disease, currently on hemodialysis

SOCIAL_HISTORY (modified)

Smoking: Former smoker, quit 12 months prior to transplantation. Alcohol: Social (1-2 drinks per month).
Occupation: High school mathematics teacher.

VITALS

Blood pressure is 132/86 mmHg

PHYSICAL_EXAM

Physical examination is unremarkable

LABS

Serum creatinine: 2.8 mg/dL (baseline: 1.2 mg/dL)
Blood urea nitrogen (BUN): 44 mg/dL
Urinalysis: No proteinuria or hematuria
Serum tacrolimus level: Within therapeutic range

Encounter 5 — 2020-06-15 — Nephrology

Encounter ID	8619	**Type**	ed
Attending	Dr. Michael Chang	**Order**	4
Department	Nephrology	**Gen. Method**	hybrid

Structured EHR Sections:

CHIEF_COMPLAINT (modified)

Malaise and decreased urine output two weeks after last clinic visit

HPI (modified)

The patient with autosomal dominant polycystic kidney disease status post deceased donor renal transplantation returns two weeks after his last clinic visit with malaise and decreased urine output. He reports mild tenderness over the graft site but no fever or chills.

PMH (modified)

Active Problem List:
- Acute rejection of lung transplant (diagnosed 2020-01-15)
- Autosomal Dominant Polycystic Kidney Disease (diagnosed 2020-03-15)
- Cytomegalovirus colitis (diagnosed 2020-04-15)
- Status post deceased donor renal transplantation
- Type 2 diabetes mellitus
- Hypertension
- Chronic kidney disease stage 5T

PSH

Undergoes kidney transplantation

MEDICATIONS (modified)

Home Medications:
- Tacrolimus 5 mg PO BID (target trough 8-10 ng/mL)
- Mycophenolate mofetil 750 mg PO BID
- Prednisone 5 mg PO daily
- Insulin glargine 24 units subcutaneous at bedtime
- Insulin lispro Sliding scale 1:12 ratio with meals
- Amlodipine 10 mg PO daily
- Trimethoprim-sulfamethoxazole Single-strength tablet PO daily (PCP prophylaxis)
- Aspirin 81 mg PO daily

ALLERGIES (modified)

NKDA

FAMILY_HISTORY (modified)

- Father: Autosomal dominant polycystic kidney disease, deceased age 58 from ruptured cerebral aneurysm
- Mother: Hypertension, type 2 diabetes mellitus
- Sister: Autosomal dominant polycystic kidney disease, currently on hemodialysis

SOCIAL_HISTORY (modified)

Smoking: Former smoker, quit 12 months prior to transplantation. Alcohol: Social (1-2 drinks per month). Occupation: High school mathematics teacher.

VITALS

His blood pressure is elevated compared with baseline.

PHYSICAL_EXAM

Physical examination shows a firm, slightly enlarged graft on deep palpation.

LABS

Serum creatinine has risen sharply from postoperative baseline values.

IMAGING

Doppler ultrasonography of the graft shows patent vessels with normal perfusion and no evidence of obstruction.

OTHER_STUDIES

A renal biopsy is performed for diagnostic evaluation.

Encounter 6 — 2021-01-15 — Pulmonology

Encounter ID	8621	**Type**	inpatient
Attending	Dr. David Kim	**Order**	5
Department	Pulmonology	**Gen. Method**	hybrid

Structured EHR Sections:

CHIEF_COMPLAINT (modified)

Progressive dyspnea, nonproductive cough, and low-grade fevers

HPI (modified)

The patient with ADPKD s/p renal transplant and prior acute rejection presents to the clinic with a 2-week history of progressive dyspnea, nonproductive cough, and low-grade fevers.

PMH (modified)

Active Problem List:
- Acute rejection of lung transplant (diagnosed 2020-01-15)
- Autosomal Dominant Polycystic Kidney Disease (diagnosed 2020-03-15)
- Cytomegalovirus colitis (diagnosed 2020-04-15)
- Chronic kidney allograft rejection (diagnosed 2020-06-15)
- Status post deceased donor renal transplantation
- Type 2 diabetes mellitus
- Hypertension
- Chronic kidney disease stage 5T

PSH

He underwent a renal transplant 6 months ago

MEDICATIONS (modified)

He has been on a regimen of tacrolimus, mycophenolate mofetil, and prednisone.

Home Medications:
- Insulin glargine 24 units subcutaneous at bedtime
- Insulin lispro Sliding scale 1:12 ratio with meals
- Amlodipine 10 mg PO daily
- Trimethoprim-sulfamethoxazole Single-strength tablet PO daily (PCP prophylaxis)
- Aspirin 81 mg PO daily

ALLERGIES (modified)

NKDA

FAMILY_HISTORY (modified)

- Father: Autosomal dominant polycystic kidney disease, deceased age 58 from ruptured cerebral aneurysm
- Mother: Hypertension, type 2 diabetes mellitus
- Sister: Autosomal dominant polycystic kidney disease, currently on hemodialysis

SOCIAL_HISTORY (modified)

Smoking: Former smoker, quit 12 months prior to transplantation. Alcohol: Social (1-2 drinks per month). Occupation: High school mathematics teacher.

VITALS

On examination, he is afebrile, tachypneic with a respiratory rate of 28 breaths per minute, and his oxygen saturation is 92% on room air.

PHYSICAL_EXAM

Lung auscultation reveals bilateral diffuse crackles.

LABS

Laboratory tests reveal:

- White blood cell count: 6,000 cells/µL
- Hemoglobin: 13.2 g/dL
- Platelet count: 250,000 cells/µL
- Serum creatinine: 1.5 mg/dL (baseline 1.3 mg/dL)
- Lactate dehydrogenase (LDH): 450 U/L (normal 140-280 U/L)

IMAGING

Chest X-ray shows bilateral interstitial infiltrates.

OTHER_STUDIES

An induced sputum sample is negative for bacterial, viral, and fungal pathogens on routine stains and cultures.

Encounter 7 — 2021-07-15 — Critical Care Medicine

Encounter ID	8622	**Type**	icu
Attending	Dr. Lisa Thompson	**Order**	6
Department	Critical Care Medicine	**Gen. Method**	hybrid

Structured EHR Sections:

CHIEF_COMPLAINT (modified)

Shortness of breath, polyuria, and increased thirst

HPI (modified)

The patient is a 46-year-old male with autosomal dominant polycystic kidney disease status post deceased donor renal transplantation and history of allograft rejection. He presents to the emergency department with a 2-day history of shortness of breath, polyuria, and an increased thirst. He reports that he has been non-compliant with his insulin regimen.

PMH (modified)

Active Problem List:
- Acute rejection of lung transplant (diagnosed 2020-01-15)
- Autosomal Dominant Polycystic Kidney Disease (diagnosed 2020-03-15)
- Cytomegalovirus colitis (diagnosed 2020-04-15)
- Chronic kidney allograft rejection (diagnosed 2020-06-15)
- History of Pneumocystis jirovecii pneumonia (diagnosed 2021-01-15)
- Status post deceased donor renal transplantation
- Type 2 diabetes mellitus
- Hypertension
- Chronic kidney disease stage 5T

PSH (modified)

- Deceased donor renal transplantation (6 months prior to initial encounter)
- Left native nephrectomy (10 years prior for infected cyst)
- Right native kidney in situ

MEDICATIONS (modified)

Home Medications:
- Tacrolimus 5 mg PO BID (target trough 8-10 ng/mL)
- Mycophenolate mofetil 750 mg PO BID
- Prednisone 5 mg PO daily
- Insulin glargine 24 units subcutaneous at bedtime
- Insulin lispro Sliding scale 1:12 ratio with meals
- Amlodipine 10 mg PO daily
- Trimethoprim-sulfamethoxazole Single-strength tablet PO daily (PCP prophylaxis)
- Aspirin 81 mg PO daily

ALLERGIES (modified)

NKDA

FAMILY_HISTORY (modified)

- Father: Autosomal dominant polycystic kidney disease, deceased age 58 from ruptured cerebral aneurysm
- Mother: Hypertension, type 2 diabetes mellitus

- Sister: Autosomal dominant polycystic kidney disease, currently on hemodialysis

SOCIAL_HISTORY (modified)

Smoking: Former smoker, quit 12 months prior to transplantation. Alcohol: Social (1-2 drinks per month). Occupation: High school mathematics teacher.

VITALS

His blood pressure is 130/85 mmHg, heart rate is 110 bpm, and temperature is 37.2°C (99°F).

PHYSICAL_EXAM

His breathing is deep and rapid, and his skin is warm and dry

LABS

- Serum glucose: 480 mg/dL
- Serum bicarbonate (HCO3): 12 mEq/L
- Arterial pH: 7.20
- Serum potassium: 5.2 mEq/L
- Serum sodium: 140 mEq/L
- Serum chloride (Cl): 95 mEq/L
- Blood urea nitrogen (BUN): 28 mg/dL
- Serum creatinine: 1.2 mg/dL
- Urine ketones: Positive

Encounter 8 — 2025-01-15 — Hematology & Oncology

Encounter ID	8624	**Type**	outpatient
Attending	Dr. Robert Garcia	**Order**	7
Department	Hematology & Oncology	**Gen. Method**	hybrid

Structured EHR Sections:

CHIEF_COMPLAINT (modified)

Fatigue, intermittent fever, and 5-pound weight loss over 3 weeks

HPI (modified)

The patient is a 50-year-old male with a deceased donor renal transplant on immunosuppression complicated by prior CMV colitis and Pneumocystis pneumonia. He presents with a 3-week history of fatigue, intermittent fever, and a 5-pound weight loss.

PMH (modified)

Active Problem List:
- Acute rejection of lung transplant (diagnosed 2020-01-15)
- Autosomal Dominant Polycystic Kidney Disease (diagnosed 2020-03-15)
- Cytomegalovirus colitis (diagnosed 2020-04-15)
- Chronic kidney allograft rejection (diagnosed 2020-06-15)
- History of Pneumocystis jirovecii pneumonia (diagnosed 2021-01-15)
- Diabetic ketoacidosis (diagnosed 2021-07-15)
- Status post deceased donor renal transplantation
- Type 2 diabetes mellitus
- Hypertension
- Chronic kidney disease stage 5T

He underwent a renal transplant 6 months ago.

PSH (modified)

- Deceased donor renal transplantation (6 months prior to initial encounter)
- Left native nephrectomy (10 years prior for infected cyst)
- Right native kidney in situ

MEDICATIONS (modified)

He has been on tacrolimus, mycophenolate, and prednisone for immunosuppression.

Home Medications:
- Mycophenolate mofetil 750 mg PO BID
- Insulin glargine 24 units subcutaneous at bedtime
- Insulin lispro Sliding scale 1:12 ratio with meals
- Amlodipine 10 mg PO daily
- Trimethoprim-sulfamethoxazole Single-strength tablet PO daily (PCP prophylaxis)
- Aspirin 81 mg PO daily

ALLERGIES (modified)

NKDA

FAMILY_HISTORY (modified)

- Father: Autosomal dominant polycystic kidney disease, deceased age 58 from ruptured cerebral aneurysm
- Mother: Hypertension, type 2 diabetes mellitus
- Sister: Autosomal dominant polycystic kidney disease, currently on hemodialysis

SOCIAL_HISTORY (modified)

Smoking: Former smoker, quit 12 months prior to transplantation. Alcohol: Social (1-2 drinks per month). Occupation: High school mathematics teacher.

VITALS

Temperature is 38.3°C (100.9°F)

PHYSICAL_EXAM

Bilateral cervical lymphadenopathy and mild hepatosplenomegaly

LABS

White blood cell count: 12,000/µL
- Hemoglobin: 10.5 g/dL
- Platelet count: 150,000/µL
- Serum creatinine: 1.2 mg/dL (baseline post-transplant)
- Lactate dehydrogenase (LDH): 450 U/L (normal 140-280 U/L)
- EBV PCR: Positive with high viral load

PATHOLOGY

An excisional biopsy of a cervical lymph node is performed, and histopathology reveals a polymorphic lymphoid proliferation with numerous plasma cells. Immunohistochemistry is positive for CD20 and EBV-encoded RNA (EBER).